

Internship Report

Telecom Customer Churn Analysis and Prediction

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github.com/Dharshinirajesh38/Customer-churn-analysis

1. Introduction

Customer churn, the loss of customers over a given period, is a critical concern for subscription-based businesses such as telecom operators, where retaining existing customers is significantly more cost-effective than acquiring new ones. This project, undertaken as part of a data analytics internship, focuses on analyzing telecom customer churn patterns and building a predictive model to identify customers at risk of leaving. Such analysis enables organizations to understand the key drivers of attrition, segment at-risk customers, and design targeted retention strategies, thereby supporting data-driven decision-making and protecting revenue.

2. Abstract

The project implements an end-to-end churn analytics pipeline by integrating SQL Server, Power BI, and Python-based machine learning. Raw telecom customer data was loaded into SQL Server, validated, cleaned, and transformed into production tables and views (vw_ChurnData, vw_JoinData) for downstream analysis. Power BI was used to build calculated columns, mapping tables, and DAX measures (Total Customers, New Joiners, Total Churn, Churn Rate), culminating in an interactive dashboard covering demographics, account and payment details, geographic distribution, service usage, and churn reasons. A Random Forest Classifier was then trained in Python on the processed dataset to predict churn, evaluated using a confusion matrix and classification report, and used to identify key churn-driving features. Predictions for newly joined customers were imported back into Power BI to create a dedicated Churn Prediction dashboard. The outcome is a complete BI-plus-ML solution that highlights major churn drivers, such as contract type and service usage, and provides a reusable framework for proactive churn management.

3. Tools Used

Category	Tools / Technologies
Database	Microsoft SQL Server (SSMS)
Visualization	Power BI (DAX measures, interactive dashboards)
Programming	Python (Jupyter Notebook, Anaconda)
Machine Learning	Scikit-learn - Random Forest Classifier
Libraries	Pandas, NumPy, Matplotlib, Seaborn, joblib, openpyxl, xlrd
Version Control	Git & GitHub

4. Steps Involved in Building the Project

- ETL in SQL Server: imported raw customer data into a staging table, performed null validation, cleaned and transformed it into a production table, and created analysis-ready views.
- Data preprocessing and transformation in Power BI: built calculated columns (Churn Status, Monthly Charge Range) and mapping tables for Age and Tenure groups; unpivoted service-related columns.
- Exploratory Data Analysis and dashboard creation: developed DAX measures and an interactive Power BI dashboard covering demographics, account/payment details, geographic churn distribution, service usage, and churn reasons with tooltips.

- Feature engineering and model building: exported SQL views to Excel, applied label encoding, and trained a Random Forest Classifier in Python to predict customer churn.
- Performance evaluation: assessed the model using a confusion matrix, classification report, and feature importance to identify key churn drivers.
- Business insights and prediction visualization: imported predicted churn results into Power BI to build a dedicated Churn Prediction dashboard segmented by demographics, account details, geography, and revenue.
- GitHub documentation and deployment: organized the project into Dataset, SQL, Power BI, and Requirements folders with a structured README and dashboard screenshots, and published it on GitHub.

5. Conclusion

The project successfully delivers an end-to-end churn analytics solution that combines relational database management, business intelligence, and machine learning. Key churn drivers, including contract type, tenure, and service usage patterns, were identified, and a working predictive model was built to flag at-risk customers for proactive retention efforts. The project strengthened practical skills in SQL-based ETL, Power BI dashboard design and DAX, Python-based machine learning with Scikit-learn, and structured project documentation using Git and GitHub. Overall, it demonstrates the ability to translate raw business data into actionable insights and reflects a workflow that can be adapted to churn or attrition problems in other industries such as retail, finance, and healthcare.